31

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE:CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I Sandhiya·K /192521378 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: K·Sadh

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%	1	1	2	1
Method Selection	20%	1	1	1	2
Solution Process	25%	2	2	1	1
Accuracy of Calculation	20%	1	1	1	2
Interpretation of Results	15%	1	1	1	0

Total Marks 29 **/40**

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2.A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the networ address, valid host IP range, and broadcast address for each subnet. Based o

the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a /27 **subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5.A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Given

Network: 192.168.20.0/24

Lab B = /27

Lab A = /26

Lab C = /28

Step 1: Lab A (/26)

* Subnet Mask: 255.255.255.192
* Network Address: 192.168.20.0
* Valid Host Range: 192.168.20.1 - 192.168.20.62
* Broadcast Address: 192.168.20.63
* Usable Hosts: 62

Step 2: Lab B (/27)

(Next available subnet after Lab A)

* Subnet Mask: 255.255.255.224
* Network address: 192.168.20.64
* Valid Host Range: 192.168.20.65 - 192.168.20.94
* Broadcast Address: 192.168.20

Step 3: Lab C (/28)

(Next available subnet after Lab

* Subnet Mask: 255.255.255.240

* Network Address: 192.168.20.96

* Valid Host Range: 192.168.20.97 - 192.168.20.110

* Broadcast Address: 192.168.20.111

* Usable Hosts: 14.

Formula:

* Usable Hosts = 2^(32 - Prefix) - 2

* /26: $2^6 - 2 = 62$

* /27: $2^5 - 2 = 30$

* /28: $2^4 - 2 = 14$.

Final Answer:

Lab	Subnet	Network address	Host range	Broadcast address	Usable Hosts
A	/26	192.168.20.0	192.168.20.1 - 192.168.20.62	192.168.20.63	62
B	/27	192.168.20.64	192.168.20.65 - 192.168.20.94	192.168.20.95	30
C	/28	192.168.20.96	192.168.20.97 - 192.168.20.	192.168.20.111	14

Given

System 1 IP: 192.168.1.130

System 2 IP: 192.168.1.190

Subnet Mask: 255.255.255.192 (/26)

formula:

Usable Hosts = $2^{(32-26)} - 2 = 2^6 - 2 = 62$

System 1

IP Address: 192.168.1.130

* Subnet Range: 192.168.1.128 - 192.168.1.191
* Network Address: 192.168.1.128
* Valid Host Range: 192.168.1.129 - 192.168.1.190
* Broadcast Address: 192.168.1.191
* IP belongs to this Subnet? yes
* usable Hosts: 62.

System 2:

IP Address: 192.168.1.190

* Subnet Range: 192.168.1.128 - 192.168.1.191
* Valid Host Range: 192.168.1.129 - 192.168.1.190
* Broadcast Address: 192.168.1.191
* IP belongs to this subnet? yes
* Usable Hosts: 62

Final Answer:

System	IP address	Network Address	Host range	Broadcast Address	Usable Hosts
System 1	192.168.1.130	192.168.1.128	192.168.1.129 - 192.168.1.190	192.168.1.191	62
System 2	192.168.1.190	192.168.1.128	192.168.1.129 - 192.168.1.190	192.168.1.191	62

Conclusion: Both 192.168.1.130 and 192.168.1.190 belong to the same subnet (192.168.1.128/26). Each /26 subnet provides 62 usable host address.

Given:

Network : 172.16.0.0/16

HR → /25

Sales → /26

IT → /27

Admin → /28

Step 1: HR Department (/25)

Subnet Mask: 255.255.255.128

Network Address: 172.16.0.0

Valid Host Range: 172.16.0.1 –
172.16.0.126

Usable Hosts: 126.

Step 2: Sales Department (/26)

Subnet Mask: 255.255.255.192

Network Address: 172.16.0.128

Valid Host Range: 172.16.0.129 -
172.16.0.190

Broadcast Address: 172.16.0.191

Usable Hosts: 62.

Step 3: IT Department (/27)

Subnet Mask: 255.255.255.224

Network Address: 172.16.0.192.

Range: 172.16.0.193 - 172.16.0.222

Broadcast Address: 172.16.0.223

Usable Hosts: 30

Step 4: Admin Department (/28)

Subnet Mask: 225.255.255.240

Network Address: 172.16.0.224

Valid Host Range: 172.16.0.225 - 172.16.0.238

Broadcast Address: 172.16.0.239

Usable Hosts: 14.

Remaining unsend IP Address Range

Start: 172.16.0.240.

End: 172.16.255.255.

Formula:

Usable Hosts = $2^{(32 - Prefix)} - 2$

$/25: 2^7 - 2 = 126$

$/26: 2^6 - 2 = 62$

$/27: 2^5 - 2 = 30$

$/28: 2^4 - 2 = 14.$

Given:

Network: 172.16.10.0/24

Subnet mask: /27 (255.255.255.224)

IP Address: 172.16.10.78

Check IP: 172.16.10.95

Step 1: Total number of students

Original prefic = /24

New prefix = /27

Number of subnets

Difference

27-24=3 bits

Subnets

$2^3=8$

Hosts per subnet

32-27=5

Address

$2^5=32$

usable

30.

IPV address

Given 2001:odb8:000:0000:0000:ff00:042:0329.

compressed form

Remove leading zeros

Replace longest consecutive zeros (::)

2001:db8::ff00:42:8329

Expanded form

2001:odb8:0000:0000:0000:ff00:0042:8329

Network prefix (/64)

first 64 bits

2001:odb8:0000:0000::/64

or

2001:~~db8~~:0:0::/64

Host Address

Hots bits

128 − 64 = 64

Total Hosts

2^{64}

= 18,446,744,073,709,

Final answer

Item	Result
Compressed IPv6	2001:db8::ff00:8329
Expanded IPv6	2001:0db8:0000:0000:000 ff00:0042:8329
Network Prefix	2001:db8:00:/64
Host Bits	64
Possible Host addresses	2^{64} = 18,446,744, 709,551,61